arXiv Submission Package & Upload Guide

Author: Rohit Ramakant Prajapati
Target Category: `math.AP` (Analysis of PDEs)
Cross-List Category: `math.LO` (Logic in Computer Science / Interactive Theorem Proving)

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1. Field-by-Field Submission Guide for arXiv.org

When logging into arXiv Submission Portal:

Title

```
On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rig
```

Authors

```
Rohit Ramakant Prajapati
```

Primary Subject Category

- Select: `Mathematics - Analysis of PDEs (math.AP)`

Secondary Subject Categories (Cross-lists)

- Select: `Mathematics - Logic (math.LO)`
- Select: `Mathematical Physics (math-ph)`

MSC-2020 Classifications

```
35Q30 (Primary), 76D05, 35B44, 35B65, 68V15 (Secondary)
```

ACM Classifications

```
F.4.1 (Mathematical Logic and Formal Languages)
```

Comments Field

```
56 pages. Includes complete open-source Lean 4 machine-checked formalization package (Mathlib 4
```

Abstract

```
 We investigate the Cauchy problem for the three-dimensional incompressible Navier-Stokes equati

First, we establish the exact orthogonal enstrophy decomposition |\nabla \omega|^2 = |\nabla |\o

Second, by optimizing dyadic Littlewood-Paley frequency projections across low and high Fourier

Third, we resolve ancient solution rigidity for the extracted limit u_\infty on (-\infty, 0] \ti

Finally, we formulate a modular, zero-sorry formal verification pipeline in the Lean 4 proof ass
```

2. Source Files to Upload

Upload `monograph.tex` directly into the arXiv file uploader. arXiv runs TeX Live on the server and will compile the `.tex` file automatically into the final PDF.